

Aiko Memory System

Implementation, Evaluation, and Comparative Analysis

A formulaic, auditable memory architecture for edge-deployed conversational AI, benchmarked against Mem0, Zep/Graphiti, and Letta/MemGPT

Aiko Memory Research Series — Paper II of II: Implementation & Evaluation

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Companion volume: “A Neuroscience-Grounded Computational Theory of Personal AI Memory” (Paper I, Version 2.0)

Source of truth: the open-source Aiko-chan repository ([cognition/memory/](#)) as of August 2026.

Abstract

This paper documents the reference engineering implementation of the memory theory developed in the companion volume (Paper I), and evaluates it against both internal benchmarks and the current (2026) landscape of agentic-memory frameworks — Mem0, Zep/Graphiti, Letta (formerly MemGPT), and related systems.

The system — codenamed Aiko — runs entirely on an 8 GB-unified-memory Jetson Orin Nano and rejects learned re-ranking in favor of closed-form, auditable scoring. The live codebase implements:

- Ebbinghaus-style exponential decay with log-compressed access strength, valence-intensity modulation of effective half-life, spaced-repetition stretch from distinct access days, and optional mood-congruent modulation;
- entity-centrality importance $I_e = (1 - \alpha)c(e) + \alpha r(e)$;
- Reciprocal Rank Fusion (RRF) across KNN, FTS5 and entity-graph signals, plus pinned, recency, access, entity-importance, context-match and negative-penalty terms;
- write-time valence/salience tagging and supersession-based belief revision;
- a nightly dream pass (salience boost, near-duplicate merge, decay prune) and monthly consolidation with an LLM merge of survivors;
- hierarchical layers including persona cache (identity facts) and scene summaries.

We document the SQLite + sqlite-vec + FTS5 schema, the full ADD/RECALL/TOUCH/dream/consolidation pipeline, the testing and monitoring harness, and end-to-end latency and footprint measurements drawn from a year-long single-deployment field test.

On that deployment the system sustained ~ 16 ms median per-turn retrieval latency, a ~ 13 MB total footprint after 1 825 stored facts, 0.72 precision@10 (vs. 0.68 for pure vector search), and $5.3\times$ monthly archival compression with no manually-identified loss of salience-tagged facts. We compare these figures, where methodologically comparable, against publicly reported 2026 benchmark data for Mem0, Zep and Letta, and discuss the engineering trade-offs — auditability and edge footprint versus leaderboard accuracy — that motivate a formulaic design.

Keywords: *agentic memory, retrieval-augmented generation, edge AI, SQLite, Reciprocal Rank Fusion, Mem0, Zep, Graphiti, Letta, MemGPT, system evaluation*

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1. Introduction & Design Principles

Paper I argued, from neuroscience and information theory, that personal-AI memory should behave as a decaying, emotionally-modulated, hierarchically-consolidated dynamical system rather than a static vector index. This paper

documents how that theory was turned into running code in the open Aiko-chan repository, and reports how the resulting system performs — in absolute terms and relative to the current field.

1.1 Why Formulas Instead of Learned Ranking

The reference deployment target is a Jetson Orin Nano with 8 GB of unified memory and a 5–15 W power envelope. Every additional resident model competes with the primary conversational LLM for the same VRAM budget; a learned-to-rank re-ranker would additionally require a labeled training pipeline and periodic retraining. The system therefore uses closed-form scoring whose principal terms are:

$$\begin{aligned}\text{strength} &= 1 + \log(1 + \min(a, A_{\text{cap}})) \\ H_{\text{eff}} &= H_0 \cdot (1 + \gamma \cdot I(v)) \cdot (1 + \gamma_s \log(1 + d)) \cdot m \\ s &= \text{strength} \cdot 0.5^{t/H_{\text{eff}}} \\ I(e) &= (1 - \alpha) c(e) + \alpha r(e) \\ \text{RRF}(m) &= \sum_i \frac{1}{k + \text{rank}_i(m)} + \text{bonuses}\end{aligned}$$

Each formula is auditable (a developer can see exactly why a memory ranked where it did), tunable by environment variables without retraining, and cheap (microsecond-scale evaluation on the Jetson).

1.2 Design Constraints

- Memory as lifecycle, not static storage — facts age; high-intensity (especially negative) facts age more slowly; spacing and mood further modulate effective half-life.
- Semantic + lexical + entity-graph fusion — a memory should be findable via any one of three independent signals, fused by RRF.
- Per-user isolation — every table is keyed by `user_id`; optional SQLCipher encryption.
- Efficient consolidation — a nightly dream pass plus a monthly consolidation gate keep the store bounded.
- Minimal context-window cost — a typical recall is formatted to well under 1.2 KB of injected context; a separate persona cache supplies stable identity facts every turn without a full search.

2. System Architecture

The live implementation lives under `cognition/memory/`. Hot-path operations — ADD, RECALL, TOUCH — run on every conversational turn. Offline maintenance comprises a nightly dream pass and a monthly consolidation job. Supporting modules include `forget.py` (decay), `entity.py` (importance, extraction, supersession, valence/salience inference), `vecstore.py` (shared sqlite-vec + RRF helpers), `schema.py` (DDL and tunables), episode/EMC distillation, and optional cross-store / spreading / neural-net overlays.

2.1 ADD (Write Path)

- Extract candidate facts from the conversational turn (LLM-assisted extraction with identity-repair post-filters).
- Embed each fact with the fixed 640-dimensional Harrier embedder (no fine-tuning).
- Check near-duplicates via KNN against existing vectors (write-dedup threshold, default ≈ 0.95).
- Classify the write as add / no-op (exact duplicate) / supersede (belief revision).
- Infer `valence_tag` / `valence_score` and `salience_hit` with cheap heuristics (and optional LLM valence override).
- Update the entity co-mention graph and enqueue the write asynchronously so it never blocks the turn.

2.2 RECALL (Read Path)

- Run a tiered quick/wide pass across three signals: KNN (semantic), FTS5 (lexical, BM25-style), and entity graph (structural / spreading).
- Fuse the rank lists via Reciprocal Rank Fusion ($k = 60$).
- Apply additive bonuses: pinned, recency, access, entity importance, mood-congruent context match; apply a soft negative-valence penalty unless the query itself is emotional/reflective.
- Optionally re-rank the high-scoring subset by recency-among-relevant.
- Touch each returned memory (increment `access_count`, update `last_accessed_at`, update `access_day_count` for spacing).
- Format the top results into a compact context block (bounded by `MEMORY_CONTEXT_TOTAL_CHARS`, default 1200). A separate persona cache (kind=`identity`) is prepended every turn without a full search.

2.3 Maintenance: Dream Pass and Monthly Consolidation

The nightly dream pass (i) boosts `access_count` on memories whose composite salience score exceeds a threshold (stored `salience_hit`, emotional intensity, access strength, spacing, mild recency prior, and keyword policy), (ii) merges near-duplicate memories discovered via KNN (retaining the higher-access / pinned copy and marking the other superseded), and (iii) prunes memories whose decayed, access-weighted score falls below the cleanup threshold after the grace period. Optional EMC episode distillation converts undistrilled episodic traces into durable semantic facts.

The monthly consolidation job applies a multi-factor retention gate (salience, novelty, spacing, connectivity / entity importance) to the past month’s daily facts, LLM-merges the survivors into compact narrative summaries, pins the result, and optionally retires the original daily facts once merged.

3. The Core Formulas (Reference Implementation)

3.1 Formula 1 — Ebbinghaus Decay with Access, Emotion, Spacing and Mood Modulators

Implemented in `cognition/memory/forget.py`:

```
strength = 1 + log1p(min(access_count, ACCESS_COUNT_CAP))
H_eff = HALF_LIFE_DAYS
        * (1 + EMOTION_GAMMA * intensity(valence))
        * (1 + SPACING_GAMMA * log1p(access_day_count))
        * mood_factor
weighted_score = strength * (0.5 ** (days_since_last_access / H_eff))
```

Intensity prefers `valence_score` $\in [-2, +2]$ mapped to $|\text{score}|/2$; falls back to tag intensities (default neg=1.0, pos=0.4, neutral=0). Mood factor lengthens half-life on valence match and shortens it on mismatch when a query valence is supplied. Defaults: $H_0 = 21$ days, cleanup threshold 0.02, grace period 35 days, $\gamma = 0.5$.

Grace-protected memories (younger than `GRACE_PERIOD_DAYS`) are never pruned.

3.2 Formula 2 — Entity Importance (Centrality + Recency)

Implemented in `cognition/memory/entity.py`:

$$I(e) = (1 - \alpha) c(e) + \alpha r(e), \quad c(e) = \frac{\text{comention}(e)}{\max_{e'} \text{comention}(e')}, \quad r(e) = e^{-\beta \cdot \Delta t(e)}$$

with defaults $\alpha = 0.4$, $\beta \approx 0.05$ per day. The map is cached briefly for recall and invalidated on writes. I_e participates in the monthly retention gate and supplies a mild additive term in recall ranking. Optional spreading activation and a neural-net overlay (Hebbian edge growth, off by default) can further propagate activation over the co-mention graph.

3.3 Formula 3 — Reciprocal Rank Fusion (Multi-Signal Fusion)

$$\text{RRF}(m) = \sum_{i \in \{\text{KNN}, \text{FTS}, \text{graph}\}} \frac{w_i}{k + \text{rank}_i(m)} + \text{recency} + \text{access} + \text{pinned} + \text{entity imp.} + \text{context match} - \text{neg penalty}$$

with $k = 60$, graph weight default 0.6, and the remaining additive weights exposed as environment variables. RRF is computed after a tiered quick/wide candidate gather so that a memory that is only moderately ranked on every signal can still surface.

3.4 Formula 4 — Emotional Imprinting and Recall Modulation

At write time, cheap lexical (and optional LLM) heuristics assign **valence_tag** $\in \{\text{neg}, \text{pos}, \text{neutral}\}$ and an integer **valence_score** $\in [-2, +2]$. These tags lengthen H_{eff} proportionally to intensity and, at recall, apply a soft proportional negative penalty unless the query itself carries matching emotional or reflective intent. Positive intensity is present but weaker than negative intensity by design (threat-asymmetric cost structure, Paper I §4).

3.5 Formula 5 — Supersession (Belief Revision)

Near-duplicate writes whose cosine similarity exceeds the dedup threshold but whose normalized text differs are classified **supersede**. The older row is marked **status='superseded'** (not deleted); the newer row records **supersedes_id**. Superseded facts are excluded from ordinary recall but remain available for history / audit, including optional chain expansion for identity/preference/plan kinds.

3.6 Formula 6 — Saliency Scoring and Consolidation Gate

Dream boost uses a composite 0...1 saliency score that combines stored **saliency_hit**, emotional intensity, log-compressed access, spacing days, mild recency, and a keyword-policy prior. Monthly consolidation retains facts above a soft multi-factor threshold (saliency + novelty + spacing + connectivity) before LLM merge into pinned narrative summaries.

4. Data Schema

4.1 memories (Core Table)

Column	Type	Notes
id	TEXT PK	UUID
user_id	TEXT	Isolation boundary
memory	TEXT	Fact string
created_at	TEXT	UTC ISO timestamp
access_count	INTEGER	Touches, capped (default 255)
last_accessed_at	TEXT	UTC ISO, or never
pinned	INTEGER	0/1; decay-proof when set
status	TEXT	active / superseded (default active)
supersedes_id	TEXT	FK to superseded memory
kind	TEXT	fact / identity / scene / plan / event / schema / ...
entities	TEXT	JSON array of entity strings
access_day_count	INTEGER	Distinct local days recalled (spacing)
valence_tag	TEXT	neg / pos / neutral
valence_score	INTEGER	-2... +2; NULL = legacy
salience_hit	INTEGER	0/1 keyword/policy flag at write
scene_id	TEXT	L2 linkage to parent scene

4.2 entity_relations and memories_vec

entity_relations stores co-mention edges (**entity_a**, **entity_b**, **relation**, **weight**, **memory_id**, **updated_at**), scoped by **user_id**. **memories_vec** stores L2-normalized 640-dimensional embeddings keyed by memory id (sqlite-vec).

5. Worked End-to-End Example

The following walkthrough traces one fact from ingestion through consolidation, illustrating how the formulas interact in the current codebase.

Day 1, 10:00 — ADD. User: “I’m starting a project called GRACE — a robotics assistant. I’ll aim to finish it by Friday.” Three facts are extracted, identity-repaired, embedded, and written (no near duplicate; all classified add). Entity edges GRACE↔robotics, GRACE↔Friday, robotics↔Friday are recorded. Valence tags are neutral; salience_hit may fire on deadline-related keywords.

Day 1, 17:00 — RECALL. Query “How’s GRACE coming along?” triggers quick/wide KNN, FTS and graph passes; RRF ranks the three GRACE-related facts at the top. Each is touched (access_count 0 → 1, last_accessed_at and access_day_count updated).

Day 8 — Decay check + new mention. After seven days without mention, strength $\times 0.5^{7/H_{\text{eff}}}$ has declined, but grace protection and any salience/access boosts keep the facts ranked highly. A new fact (“Oppa completed GRACE ahead of the Friday deadline”) is added (similarity below supersession threshold $\Rightarrow$ add).

Night of Day 8 — Dream pass. Composite salience elevates the GRACE cluster; access_count is boosted. Near-duplicate merge (if any) retains the higher-access copy. Scores remain above the cleanup threshold, so nothing is pruned.

Month later — Monthly consolidation. The multi-factor gate keeps the high-connectivity, deadline-bearing facts; LLM merge produces a compact pinned monthly summary and the atomic daily rows may be retired.

6. Testing, Monitoring & Tuning Harness

Five production metrics gate health:

- **Retention rate** — fraction of day-1-accessed memories still retrievable after 30 days (targets $\geq 70\%$ for salience-tagged, $\geq 50\%$ for routine).
- **No premature forgetting** — zero cleanups of facts younger than the grace period.
- **Entity importance rank stability** — top-10 entities drop at most two ranks over 30 days.
- **RRF coverage** — graph-only rescues appear in a non-trivial fraction of queries.
- **Consolidation fidelity** — zero critical facts lost; compression ratio in a healthy band.

Diagnostic playbook (symptoms $\rightarrow$ likely cause $\rightarrow$ fix) follows the same pattern as v1: lengthen half-life or raise cleanup threshold if decay is too aggressive; verify graph weight and entity extraction if the graph signal is unused; re-run maintenance and inspect dedup threshold on duplicate surfacing; raise consolidation soft threshold / max-month budget if over-deletion occurs.

7. Performance Evaluation

All figures are drawn from the same year-long, single-deployment field test (Jetson Orin Nano, 8 GB unified memory) documented in Paper I, plus system-level instrumentation. As an $N = 1$ field deployment these numbers characterize one system under one usage pattern.

7.1 Latency Budget

Total per-turn context-injection latency — persona cache (cached) + KNN + FTS + graph + RRF scoring + touch metadata + formatting — sums to approximately 16 ms median, comfortably inside a < 100 ms conversational budget. Heavy operations (embedding calls, dream merge, monthly LLM merge) remain off the hot path.

7.2 Storage Footprint

After $\sim 1\,825$ facts the on-disk footprint of memories, 640-d vectors, entity graph, FTS5 index and ancillary tables is approximately 13 MB — well under 1 % of the Jetson’s 8 GB envelope. The practical constraint remains the LLM context window, not disk.

7.3 Retrieval Quality

On a 100-query human-judged sample, RRF fusion improved precision@10 from 0.68 (pure KNN) to 0.72 and recall@10 from 0.61 to 0.68, while surfacing an additional share of results that neither KNN nor FTS alone returned in their top candidates (graph rescues).

7.4 Consolidation Efficiency

Monthly compression (pre-LLM-merge survival $\approx 54.7\%$, post-merge overall $\approx 5.3\times$) with zero critical facts lost under manual review of dropped rows. Persona cache remains a handful of stable identity facts updated infrequently.

8. Comparative Analysis vs. Established Agentic Memory Systems

By mid-2026 the agentic-memory space has consolidated around a handful of widely adopted frameworks. Cross-system numbers should be read cautiously: different projects report different harnesses, and third-party round-ups are not a single controlled study.

System	Core approach	Ranking	Typical footprint
Mem0	Drop-in memory API; LLM extraction over a vector store	Embedding + heuristics	Hosted or self-hosted SDK
Zep / Graphiti	Temporal knowledge graph	Graph traversal + hybrid scoring	Graph DB; heavier self-hosting

System	Core approach	Ranking	Typical footprint
Letta (MemGPT)	OS-inspired agent runtime (RAM / disk)	Agent-directed paging	Full agent runtime
Aiko (this work)	Formulaic decay/consolidation over SQLite + sqlite-vec + FTS5	Closed-form RRF + modulators, no learned weights	~13 MB/year; full edge on 8 GB device

Aiko occupies the niche of hard edge budget + formula-level auditability and emotional/psychological realism in retention. It does not claim to match graph-reasoning depth (Zep) or self-managing runtime sophistication (Letta) on shared benchmarks it has not yet run. The choice among these systems remains an architectural commitment, not a checkbox.

9. Key Engineering Decisions & Trade-offs

- **No learned ranking.** Formulas stay inside the 8 GB budget, need no training pipeline, and remain inspectable. Trade-off: possible sub-optimality on edge cases versus a well-trained re-ranker.
- **Supersession instead of deletion.** Preserves belief history for audit without an LLM contradiction pass. Trade-off: modest extra store growth; superseded rows excluded from normal recall.
- **Entity graph + optional spreading.** Rescues facts that are only findable through relationships. Trade-off: depends on reliable (currently rule-based) entity extraction.
- **Separate monthly consolidation.** Nightly dream handles duplicates and ordinary decay; monthly compression handles volume. Trade-off: monthly facts are lossy by design.
- **Log-compressed access strength.** Prevents heavily-touched memories from becoming effectively immortal while still rewarding spaced retrieval.

10. Limitations & Future Work

- Entity extraction remains largely rule-based; a lightweight NER or embedding-based linker would raise graph rescue rates.
- Valence estimation is heuristic (with optional LLM override); sarcasm and implicit emotion remain imperfect.

- Single-user validation ($N = 1$); multi-user isolation is already implemented, but population-level claims require multi-subject study.
- Decay parameters are global defaults; per-user Bayesian tuning is a natural extension.
- Aiko has not yet been evaluated on LongMemEval or Deep Memory Retrieval; running those shared benchmarks is the highest-priority next step for quantitative comparison.
- Optional neural-net overlay and full spreading activation are present but off or lightly weighted by default; their contribution under sustained multi-year load is still being measured.

11. Conclusion

Aiko’s memory system turns the theoretical account of Paper I into a working, fully auditable pipeline built on SQLite, sqlite-vec and FTS5, running comfortably inside an 8 GB edge-hardware budget. The closed-form decay, importance, RRF, emotional, supersession and salience formulas work together to give a system that is tunable without retraining, explainable at every ranking decision, and lean enough to run indefinitely on a single-board computer.

Measured against itself the system meets its design targets: ~ 16 ms per-turn retrieval, ~ 13 MB/year storage growth, a measurable precision gain from multi-signal fusion, and $5.3\times$ monthly compression without loss of load-bearing facts. Measured against the broader 2026 landscape it occupies a distinct niche complementary to general-purpose, benchmark-optimized memory services.

References

- Cormack, G. V., Smucker, M. D., & Clarke, C. L. (2009). Efficient and effective spam filtering and re-ranking. *Proceedings of SIGIR 2009*, 713–714.
- Robertson, S., & Walker, S. (1994). Some simple effective approximations to the 2-Poisson model. *Proceedings of SIGIR 1994*, 232–241.
- Packer, C., et al. (2023). MemGPT: Towards LLMs as operating systems. *arXiv:2310.08560*.
- Rasmussen, P., et al. (2024–2025). Zep: A temporal knowledge graph architecture for agent memory (Graphiti). Zep technical report.
- Mem0 project documentation, mem0.ai (accessed 2026).
- Letta (formerly MemGPT) project documentation, letta.com (accessed 2026).
- Aiko-chan source repository, github.com/OppaAI/Aiko-chan (dev branch, August 2026) — especially `cognition/memory/{forget,entity,schema,vecstore,episode_dream}.py`.

Appendix A: Configuration Reference

Parameter	Default	Range / notes	Interpretation
FORGET_HALF_LIFE_DAYS	31.0	[7, 60]	Base half-life (days)
FORGET_CLEANUP_THRESHOLD	0.02	[0.01, 0.1]	Score floor for prune
FORGET_GRACE_PERIOD_DAYS	35	[14, 60]	Protect new facts
FORGET_ACCESS_COUNT_CAP	255	—	Cap for log1p strength
FORGET_EMOTION_GAMMA	0.5	[0, 1]	Valence stretch of H_{eff}
FORGET_INTENSITY_NEG / POS / NEUTRAL	1.0 / 0.4 / 0.0	—	Tag intensities
FORGET_MOOD_MATCH_SLOWDOWN	1.0	—	Match lengthens H_{eff}
FORGET_MOOD_MISMATCH_ACCELERATION	0.3	—	Mismatch shortens H_{eff}
RRF_K	60	[30, 100]	RRF damping constant
MEMORY_RANK_GRAPH_WEIGHT	0.6	[0, 1]	Graph contribution
ENTITY_IMPORTANCE_ALPHA	0.4	[0.2, 0.6]	Recency vs centrality
MEMORY_RANK_ENTITY_IMPORTANCE_WEIGHT	0.008	—	Mild recall boost
WRITE_DEDUP / DREAM_MERGE thresholds	≈ 0.95 / 0.88	—	Supersede / merge
CONSOLIDATION soft threshold / max month	≈ 0.4 / 30	—	Monthly gate
EMBED_DIMS	640	—	Harrier embedding dimension